

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 6: Where Does a Maize or Bean Plant Grow? Measuring Growth and Regions of Growth in Seedlings

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
B	D	A	A	B	D	D	A

USE AFTER: PREDICT PHASE — Predict: individual drawing and Think-Pair-Share on where growth in length takes place on the seedling

Q1. A farmer in Nakuru plants maize and beans. A student predicts where a young root becomes longer. Which prediction best matches how growth in length happens in a root?

- A. Growth happens only at the part of the root closest to the seed
- B. Growth happens mainly in a region near the root tip, where cells divide and elongate**
- C. Growth happens evenly along the whole length of the root
- D. Growth happens only in the oldest part of the root, far from the tip

Correct answer: B. Growth in length is concentrated in the root tip region (apical meristem and the zone of elongation behind it), not evenly along the root.

USE AFTER: OBSERVE PHASE — Observe: marking maize and bean roots with ink at 5 mm intervals and re-measuring after 24 and 48 hours

Q2. Learners mark a young bean root with ink every 5 mm starting from the tip, then return it to moist soil. After 48 hours they measure the distances between the marks. Which marks will have moved furthest apart?

- A. All marks will have moved apart by the same amount
- B. The marks in the oldest part of the root, furthest from the tip
- C. None, because roots only grow in width
- D. The marks nearest the root tip**

Correct answer: D. Cells near the root tip divide and elongate, so the spaces between marks there increase most. Marks higher up stay almost the same distance apart.

USE AFTER: OBSERVE PHASE — Observe: marking maize and bean roots and shoots at equal intervals, and the teacher question 'Why do we mark at equal distances?'

Q3. In the seedling investigation, why do learners mark the root at equal intervals (for example every 5 mm) from the tip?

- A. So that any change in the distance between marks shows which region has grown**
- B. So that the ink dries faster on the root
- C. So that the root grows straighter
- D. So that the seedling grows faster

Correct answer: A. Equal starting spacing makes a fair comparison: if one gap becomes larger than the others, growth happened in that region.

USE AFTER: OBSERVE PHASE — Observe: plotting the maize and bean growth curves from the height-against-time data table

Q4. A maize plant was 4 cm tall on day 7 and 32 cm tall on day 21. What was its average growth rate between day 7 and day 21?

- A. 2 cm per day**
- B. 1.5 cm per day
- C. 4 cm per day
- D. 28 cm per day

Correct answer: A. Growth = $32 - 4 = 28$ cm over $21 - 7 = 14$ days. Rate = $28 \div 14 = 2$ cm per day.

USE AFTER: EXPLAIN PHASE — Explain: CER discussion of the S-shaped growth curve and the oral probe 'What does the steep part of the curve show?'

Q5. On an S-shaped growth curve of a maize plant (height against time), what does the steep middle part show?

- A. Growth has stopped because the plant is mature
- B. Growth is rapid, with active cell division and photosynthesis**
- C. Growth is slowing because resources are limited
- D. The seed is still using stored food and growth is slow

Correct answer: B. The steep part is the rapid (log) phase, when cells divide actively and photosynthesis supports fast increase in height.

USE AFTER: EXPLAIN PHASE — Explain: Claim-Evidence-Reasoning (CER) frame 'I claim that _____. My evidence is _____. This is because _____.'

Q6. A group is writing a Claim-Evidence-Reasoning statement about where a root grows. Which of these is a piece of EVIDENCE?

- A. Growth in length is concentrated at the root tip.
- B. All plants grow in the same way.
- C. This happens because cells in the meristem divide and then elongate.
- D. The marks nearest the root tip moved 8 mm apart in 48 hours, while marks higher up stayed the same distance apart.**

Correct answer: D. Evidence is measured data from the investigation. The first choice is a claim, the third is reasoning, and the last is an unsupported statement.

USE AFTER: DQB PHASE — Explain and DQB: key terms on the board and the misconception 'growth is not only height'

Q7. Which statement best defines growth in a plant?

- A. Any change in the shape of the plant, with no change in size
- B. A temporary increase in size caused by a seed absorbing water
- C. An increase in height only
- D. A permanent increase in size and dry mass**

Correct answer: D. Growth is a permanent increase in size and dry mass. Swelling from water is temporary, and growth can be measured as height, length, width or dry mass, not height only.

USE AFTER: MODEL BUILDING PHASE — Model: revising the poster model of primary growth (apical meristems) and secondary growth (vascular cambium in the bean stem)

Q8. Which plant can become thicker through secondary growth, and which tissue is responsible?

- A. Bean, through the vascular cambium**
- B. Maize, through the vascular cambium
- C. Maize, through the apical meristem at the shoot tip
- D. Bean, through the apical meristem at the root tip

Correct answer: A. Bean is a dicot and can thicken its stem because of the vascular cambium (a lateral meristem). Maize is a monocot with no secondary growth, and apical meristems cause growth in length.